Split-graph secant complete-cost panel: results (attempt 3)

Verdict (PROTOCOL.md section 9, applied mechanically): REJECTED at the tested scale for both candidates, cs1 (secant, four solves) and cs2 (secant, two epsilon-shared solves), additionally labelled UNRANKED under section 5a. [pcc:unverified]

cs1 vs same-base dense cd (field ops, excl. construct): 4 cells higher, 2 indistinct, 0 lower, 0 advancing. cs2: 6 of 6 higher (GM 3.59 to 4.44), 0 advancing. Handled degree falls from 2d+2 to d+1 on every unit; the complete same-base field-operation cost per independent relation does not fall. With native work priced at 1 per operation the order flips (cs1: 6/6 cells, cs2: 5/6), so no ranking in either direction except cs2 in C1 (higher across the whole 0/0.1/1 band).

Gates (section 8): exactness PASS (7380 rows, 0 mismatches/replay failures/unresolved/non-zero-dim/censored/wrong lambda); self ratio PASS (tautological as implemented); positive control PASS (pc2x solve-phase field ops exactly 2x on 30/30 units in every cell); A/A hs2/hs2b CI overlaps the +-5% band in all 6 cells (both scopes); null ld/hd CIs all contain 1. Clean cells: 6 of 6.

Statistics: geometric mean of per-unit cost ratios, 4000-resample percentile bootstrap over the 120 paired units per cell, margin 5%. The analysis reproduced the instrument's own bootstrap from records.jsonl: 204 summary.json rows, max deviation 5e-7, 0 verdict mismatches; an independent bootstrap gives the same verdict on every decision row. Card A3-V01 recheck: 600 GM rows, 0 mismatches.

Identity: PROTOCOL.md sha256 20acbf94... equals the frozen value and predates the run; native/source lock OK (9 files); image isolab/tasks@sha256:d1ab2c5c44bc..., source f13a690df016; binary sha256 dc7c013bc2ac...; SHA256SUMS OK on the 4 panel files; 7380 rows as expected.

Execution: attempts 1 (sgsecant-panel-20261107) and 2 (sgsecant-panel-20261108-a2) were never scheduled (DeadlineExceeded, no data). Attempt 3 (job uid 517e76f8...) ran 10:46:31-10:51:26Z on 2026-11-08, exit 0; the Pod object was deleted with its node, so the pod and exit code were taken from the EKS API-server audit log (one job-controller create owned by the job uid, one node-reported termination, exit 0) by card A3-V01.

Scope: exact operation counts of one counted Buchberger+FGLM engine on toy prime fields (p 151..991), arity 2; no timing claim; distance to rho uses the internal rho reference only (not the matched ecbench rho); not a leaderboard row.

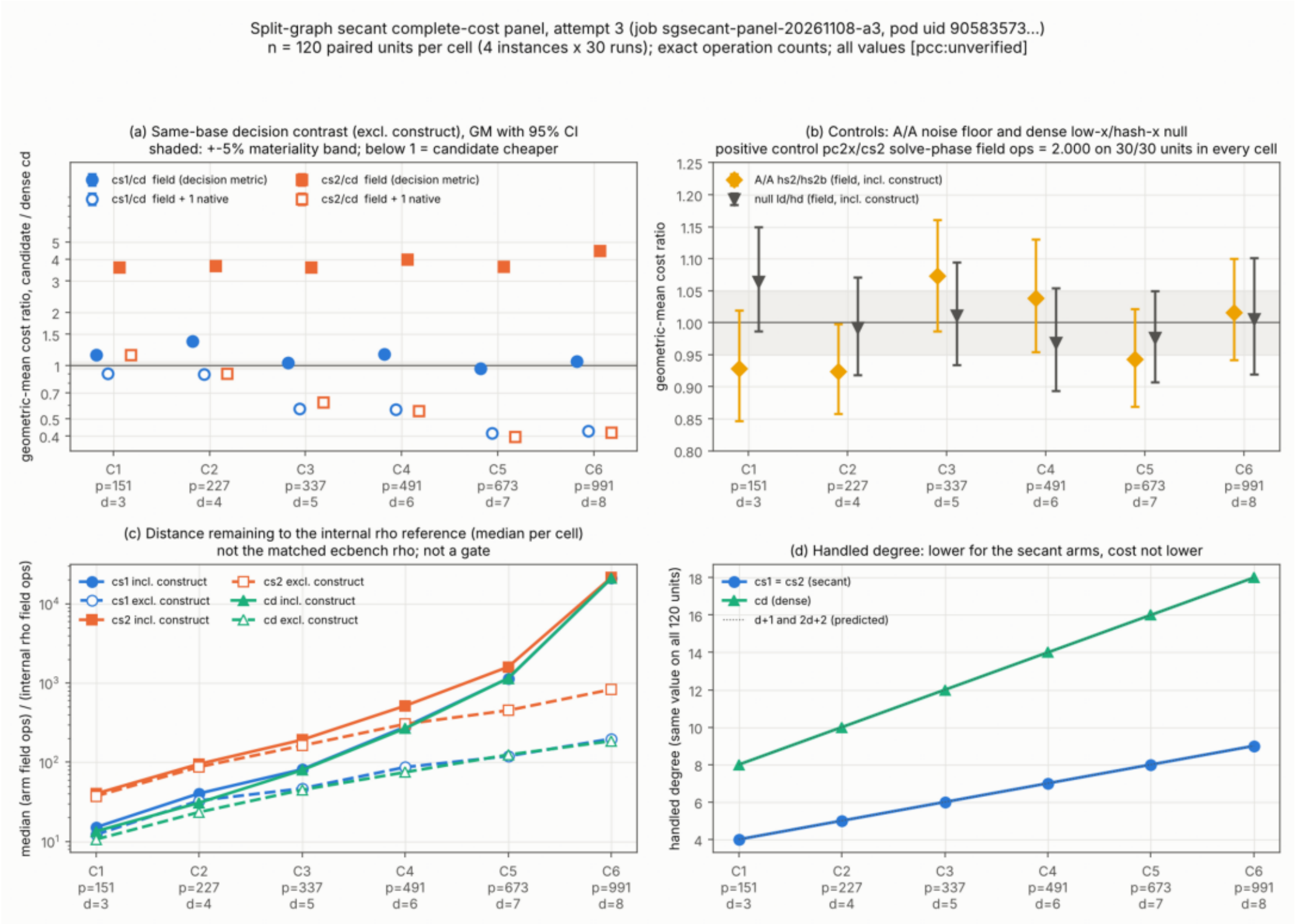


Figure 1. (a) same-base decision contrasts, GM with 95% CI; (b) A/A and null controls; (c) median distance to the internal rho reference; (d) handled degree. Source: results/A3/analysis/a3sgsecant_summary.json.

Per-cell verdicts, intervals and controls

n = 120 paired units per cell (4 instances x 30 runs; pc2× 30 units, instance 0). Verdict: lower iff CI upper < 0.95, higher iff CI lower > 1.05. Effects smaller than the cell's A/A half-width are not claimed (section 8.4): cs1/cd field in C3, C5, C6 and cs1/cd +0.1 native in C1.

cs1/cd same base, excl. construct (decision scope)

cell	p	d	field GM [95% CI]	verdict	mul	inv10	+0.1nat	+1nat GM	+1nat
C1	151	3	1.149 [1.138,1.160]	higher	higher	higher	higher	0.898	lower
C2	227	4	1.368 [1.359,1.377]	higher	higher	higher	higher	0.892	lower
C3	337	5	1.034 [1.030,1.038]	indistinct	higher	higher	lower	0.571	lower
C4	491	6	1.157 [1.151,1.162]	higher	higher	higher	lower	0.567	lower
C5	673	7	0.958 [0.955,0.961]	indistinct	indistinct	indistinct	lower	0.413	lower
C6	991	8	1.060 [1.057,1.063]	higher	higher	higher	lower	0.426	lower

-> clean cells 6, advancing 0, field higher 4, not lower 6: REJECTED + UNRANKED

cs2/cd same base, excl. construct (decision scope)

cell	p	d	field GM [95% CI]	verdict	mul	inv10	+0.1nat	+1nat GM	+1nat
C1	151	3	3.592 [3.523,3.660]	higher	higher	higher	higher	1.146	higher
C2	227	4	3.648 [3.599,3.694]	higher	higher	higher	higher	0.898	lower
C3	337	5	3.586 [3.548,3.622]	higher	higher	higher	higher	0.620	lower
C4	491	6	3.981 [3.936,4.022]	higher	higher	higher	higher	0.557	lower
C5	673	7	3.639 [3.611,3.666]	higher	higher	higher	higher	0.395	lower
C6	991	8	4.443 [4.416,4.470]	higher	higher	higher	higher	0.418	lower

-> clean cells 6, advancing 0, field higher 6, not lower 6: REJECTED + UNRANKED

Controls (field, incl. construct)

cell	A/A hs2/hs2b GM [CI]	half-width	ld/hd GM [CI]	pc2x 2x
C1	0.928 [0.846,1.019]	0.087	1.065 [0.987,1.150]	30/30
C2	0.924 [0.858,0.998]	0.070	0.992 [0.919,1.071]	30/30
C3	1.073 [0.987,1.161]	0.087	1.011 [0.934,1.095]	30/30
C4	1.039 [0.955,1.130]	0.088	0.969 [0.893,1.055]	30/30
C5	0.943 [0.870,1.022]	0.076	0.977 [0.908,1.049]	30/30
C6	1.016 [0.942,1.100]	0.079	1.006 [0.920,1.101]	30/30

Median distance to the internal rho reference (field, incl. construct)

cell	cs1	cs2	cd
C1	15.1	40.1	13.4
C2	40.3	94.9	30.7
C3	81.1	191.4	79.3
C4	279.0	513.9	266.7
C5	1139.0	1590.6	1144.0
C6	20837.0	21343.3	20830.0

Pre-registered expectations (PROTOCOL section 10) and caveats

E1 extractor exact: HELD (0 mismatches on 7380 rows).

E2 handled degree d+1 (secant) vs 2d+2 (dense): HELD on every unit of every cell.

E3 cs1/cd field 'about 1.0 to 1.4x', at or above cd: GM 1.149, 1.368, 1.034, 1.157, 0.958, 1.060. Held in 5/6 as a literal range; C5 (0.958) below it but indistinct, never lower.

E4 cs2/cd 'about 2 to 4x': GM 3.59, 3.65, 3.59, 3.98, 3.64, 4.44. Held in 5/6; C6 above 4.

E5 +1 native flips the order in several cells: HELD (cs1 lower in 6/6, cs2 in 5/6).

E6 expected outcome REJECTED with UNRANKED order: HELD for both candidates.

Caveats. The bootstrap resamples paired units, not the 4 instances, so between-instance variance is not in the CIs. The A/A contrast is cross-base; its half-width (7-9%) is about ten times the same-base paired CIs, and in C2 its CI excludes 1. The self-ratio gate divides a record by itself. summary.txt labels cs2/cd 'PRIMARY' although DESIGN.md calls E2 not the primary; section 9 treats both candidates separately, so this does not affect the verdict. Native operations are an engine-specific proxy. The distance to rho uses the internal reference, not the matched ecbench rho with the 5a band.

UNRESOLVED: none affecting the verdict. Ambiguities checked (A/A metric and scope, 'order differs' reading, bootstrap RNG, half-width scale, 'far from 1') each gave one answer under every reading.

Analysis scripts are kept out of the repository (AGENTS.md: no Python for result generation) and archived to S3 by sha256 (results/A3/analysis/scripts-archive.json). Full report: research/split_graph_secant_cost_20261107/RESULTS.md.